"Soil"

Soil Resources

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The soil is a mixture of mineral grains, organic material, water, and gas. The mineral grains include clay, silt, sand, and rock fragments.

- Clay is so fine grained and closely packed that water and air do not flow readily through sandy soil
- Loam is the most fertile soil, a mixture especially rich in sand and silt with generous amounts of organic matter
- Humus is an essential component of most fertile soils. If you walk through a forest or prairie, you can find bits of organic litter-leaves, stems, and flowers that have not decomposed- on the soil surface. When this litter decomposes sufficiently that you can no longer determine the origin of individual pieces, it become humus.

Humus and Plant Growth

Humus soaks up so much moisture that humus-rich soil swells after a rain and shrinks during dry spells. This alternate shrinking and swelling loosens the soil, allowing roots to grow into it easily. A rich layer of humus also insulates the soil from excessive heat and cold and reduces water loss from evaporation. Humus enriches the soil in nutrients and makes them available to plants. In intensive agriculture, farmers commonly plow the soil and leave it exposed for weeks or months. Humus oxidizes in air and decomposes; at the same time, rain dissolves soil nutrients and carries them away.

Farmers replace the lost nutrients with chemical fertilizers but rarely replenish the humus. As a result, much of the soil's ability to absorb and regulate water and nutrients is lost. When rainwater flows over the surface, it transports soil, particles, excess fertilizers, and pesticide residues, polluting streams and ground water.

Land Use and Abuse

The land is a major constituent of the lithosphere and is the source of the many materials essential to man and other organisms. It forms about $1/5^{th}$ of the Earth's surface, covering approximately 13,393 million hectares.

Land Use:

Most human or natural activities need space for their location and development. This space is provided for, by land, which is put to various uses like food and energy production, waste disposal, industrial, commercial and residential purposes.

Impact of Land-use on Environmental Quality

Land use involves economic activities which in turn lead to several environmental problems like:

a. Problem of dealing with the pollutants discharged
b. Problem of disposal wastes
c. Consumption of natural resources for the economic activities
d. Malfunctioning of the ecosystem by disturbing the ecological cycles and wild life habitats.

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Use of agricultural land has become increasingly specialized owing to the changes occurring in agricultural practices. An environmental resource such as water is used up for irrigation.
Often, there are changes within a particular land use category. These activities often result in major changes in the landscape, thereby reducing its capacity to support a diversified and balanced wild-life, reducing its tourist attraction and its recreational value. As a consequence, the potential for multiple uses of these areas gets highly reduced.
Changes also result from one land use to another. For instance,

a. Agricultural land used as an urban area
b. Agricultural land used for forestry
c. Forest area cleared for use as agricultural land

These changes are usually made to achieve better economic use of the land resource from the owner's point of view.
Sometimes many environmental conflicts arise between adjacent land users, for instance,

a. Residential areas located near industrial areas are affected by air pollution due to effluents of energy and industrial facilities.
b. Residential areas located near airports or along highways and motor ways suffer nuisance of noise pollution.
c. Intensive stock breeding units are the source of offensive odors of nearby dwellers.
d. Development of linear infrastructure like roads and railway tracks in rural areas affects the existing land use as well as natural ecosystems like forest or wetlands.

Land- degradation

The surface layer of land is called *soil*. It covers about four-fifths (4/5) of the land areas. Fertility or the productive capacity of the soil depends on the minerals it contains. As the minerals abound in the top layer of the soil, so the top layer is best for vegetation. Land degradation refers to the deterioration or loss of fertility or productive capacity of the soil. The factors that deplete the mineral content of the top layer or damage it reduce the fertility of the soil. These factors are the following:

a. Soil erosion
b. Soil pollution
c. Salination and water logging
d. Shifting cultivation
e. Desertification
f. Urbanization

Soil erosion is the loss or removal of the superficial layer of the soil by the action of water, wind, or by activities of man. The factors influencing the extent to which soil erosion will occur are:

a. Distribution, intensity, and amount of rainfall

Unequal distribution of rainfall during the year results in heavy rainfall being restricted to a few months of the year. The soil is unable to absorb this heavy rainfall and, therefore, there is plenty of run-off water which removes layers of soil as it moves along, resulting in soil erosion.

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b. Slope of the ground

If the land has steep slopes then infiltration or rain water decreases and the run off is much faster resulting in more soil erosion.

c. Nature of the soil

If equal slopes are exposed to the same type of rainfall than light, open soils lose more silt than heavier loams, which merely swell up by wetting.

d. Vegetation Cover

Vegetation holds the soil in place by forming a network of roots of the plants.

Rain falling on thick vegetation cover gets partly soaked into the ground, is partly absorbed by the vegetation and partly evaporates. Soil erosion is negligible due to less surface run-off. On the other hand, rain falling on bare land causes soil erosion as the top soil is loose and is easily carried away by rain water.

e. Soil Mismanagement

Faulty methods of surface drainage, over grazing, wrong methods of cultivation (cutting fields along the direction of hill slopes), forest-fires, and removal of forest litter are common practices that aggravate the erosion of soil.

f. Soil Pollution

Soil pollution is the reduction in the productivity due to the presence of soil pollutants.

Organic insecticides like DDT are used against soil-borne pests. They accumulate in the soil as they grade very slowly by soil and water bacteria. Consequently, they have a very delirious effect on the plant growth- stunting their growth and reducing the yield and size of the fruit.

The degradation products of these insecticides may be absorbed by the plants from where they reach the animals and man through the food chains.

Radioactive wastes from mining and nuclear processes may reach the soil via water or as "fall-out". From the soil they reach the plants and then into the grazing animals (livestock) from where ultimately reach man through milk and meat etc. resulting in retarded and abnormal growth of man.

Human and animal excreta used as organic manure to promote crop yield, pollute the soil by contaminating the soil and vegetable crops with the pathogens that may be present in the excreta.

Salination and Water Logging

Salination is the increase in the concentration of soluble salts in the soil. In arid areas with low rainfall, poor drainage, and high temperatures water evaporates quickly leaving behind the salts in high concentrations.

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Salination results from two (2) processes:

a. Due to poor drainage of irrigation and flood waters, the salts dissolved in these waters accumulate on the soil surface.
b. In summers, the salts from deeper strata are drawn up by capillary action and get deposited on the surface. Excess of these salts form a white crust on the soil surface and are injurious to the survival of plants. The water absorption process of the plant is severely affected.

Though sufficient water is present in the soil, it is pathologically unavailable to the plants because of the higher concentrations of the soil solution.

Water can be absorbed in the plant only if its concentration is higher than the water concentration inside the plant.

Shifting Cultivation

This is a traditional, sustainable method of agriculture that has been practiced by indigenous tribes for centuries. Along with other aspects of their culture and traditional way of life, it is under threat from large-scale clearance of the forests.

- A small area of land is cleared and the vegetation burned, providing a source of nutrients from the ash.
- For a few years the soil remains sufficiently fertile for the tribe to grow crops.
- When the soil's fertility is exhausted, the tribe moves on and clears another small area of forests.
- The original area is regenerated, as it receives nutrients and seeds from surrounding vegetation.
- It is sometimes called "slash-and-burn"

The Kaingin-making is a farming system based on shifting, or slash-and-burn clearing of forest for the planting of agricultural and agro-forestry crops.

Kaingin is a portion of the forest land, whether occupied or not which is subjected to shifting and /or permanent slash-and-burn cultivation having little or no provision to prevent soil erosion.

Desertification

Increasing human population has put a great pressure on the land. Vast areas of land have been cleared for cultivation of agricultural crops. The destruction of natural vegetation results in accelerated soil erosion due to the removal of the vegetation cover. Erosion of the top fertile soil results in loss of soil productivity and formation of deserts. The wind causes the shifting of sand dunes from one place to another, increasing the proportion of fertile land getting converted into deserts each year.

Excessive grazing (to put livestock out to feed) is another factor resulting in desertification. All efforts to check the advancement of deserts by afforestation are of no use because large herds of goats and other animals destroy the samplings planted. Many deserts in the world are a result of human activities.

Urbanization

Productive area is fast reducing because of developmental activities such as human settlements, industries, roads, railways, airports, dams, canals, etc.

The balance of nature has been upset, exposing the soil over millions of acres of land to wind and rain, which in turn have reduced mountains to hills and hills to plains. The loss cannot be compensated through increased productivity or economic progress.

Control of Land Degradation

The process of soil formation is so slow that soil can be considered as a non-renewable resource. Therefore, control of land degradation is very important and can be achieved by taking the following measures:

1. Afforestation and Reforestation

Afforestation means growing forests where there were no forests before, may be due to lack of seed trees or adverse factors.

Reforestation means replanting forests at places where they were destroyed by overgrazing, excessive felling, forest fires etc. At such places natural regeneration is not possible and artificial planting of trees is necessary.

2. Better Agricultural Practices

a. Terracing: A large sloping drainage area is divided into a number down the speed or run-off water (therefore, control soil erosion), hold the water longer on the land so that more of it is soaked into the land and this results in higher crop yields.
b. Regular Cultivation: Fields should not be left bare and there should be regular cultivation of the fields with a vegetation cover so that erosion is checked and maximum water infilters into the soil due to minimum run-off of water.
c. Crop Rotation: This involves sowing of different crops in the same field in a regular sequence for a number of years.

   Nutrient consuming crops like peas, which enrich the depleted soil with nitrates. In the following year, some other nutrient-consuming crops (Wheat, Barley) can be grown in the same field
d. Fallowing: Aeration, texture, and fertility of the soil can be improved by ploughing the fields extensively and then leaving them without sowing for a year or two (2).

3. Planting Wind Breaks and Shelter Belts

To check the velocity of wind, two (2) or more rows of tall trees are planted at right angles to the direction of the prevailing winds. These wind breaks check the movement of wind and therefore, check soil erosion.

Landscape Pollution and Solid Waste Management

Human and animal activities generate many wastes that are discarded as useless or unwanted. These wastes are normally solid and result in landscape pollution. The term solid wastes encompasses the highly heterogeneous mass of discarded

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materials or throwaways from the urban community (i.e. residential and commercial activities) as well as the more homogeneous accumulation of wastes generated by agricultural and industrial activities.

Landscape pollution is the third important pollution after air and water pollution.

Sources and Classification of Solid Wastes

Solid wastes can be classified into the following categories:

a. *Garbage or Food-Wastes.* These are meat, fruit, or vegetable residues that decompose rapidly, especially in warm weather
   Examples: Vegetable and fruit peels, leftovers, meat, bones, spoiled food items
b. *Rubbish*. These do not decompose rapidly. It has two (2) types:
   i. Combustible- Example: paper, cardboard, textiles, wood items, rubber, leather, plastic containers
   ii. Non-combustible- Example: crockery, metals, aluminum cans, tin cans, empty glass bottles
c. *Agricultural Wastes*. These include crop residues from agricultural fields, farm manure, etc.
   Examples: Cotton, rubber, tea, coffee, coconut, sugarcane wastes, rice straw, cattle-shed wastes
d. *Industrial Wastes*. These arise from industrial activities.
   Example: Flyash, sewage, chemicals, sludge, paints, toxic metals, etc.
e. *Hazardous Wastes*. These include radioactive wastes, toxic chemicals, flammable wastes, and explosives, hazardous biological wastes from hospitals or research institutions.
f. *Pathological Wastes*. These include carcass of animals, slaughter house wastes (blood, pieces of meat, hair, fat, bone chippings, hides, etc.)
g. *Demolition and Construction Wastes*. These include stones, bricks, concrete, dust, plaster, electrical, plumbing, and sanitary parts
h. *Aquatic Weeds*. These are a menace because of their prolific growth. They increase water-borne diseases, hinder traffic and fishing and cause eutrophication.
i. *Miscellaneous Wastes*. Wastes not included in any of the above categories.
   Examples: Street sweepings, roadside litter, dead stray animals, abandoned vehicles etc.

The main sources of solid wastes are domestic, commercial, industrial, municipal, and agricultural wastes.

Agriculture and the food processing industry are considered to be the largest contributors to the total annual production of solid wastes. Control techniques for air and water pollution also result in waste production for instance,

- Sewage treatment plants result in sludge
- Fly ash is recovered from flue gases,
- Farm animal wastes may leach into the water supplies causing water pollution,
- Decaying of organic matter produces obnoxious odours causing air pollution.

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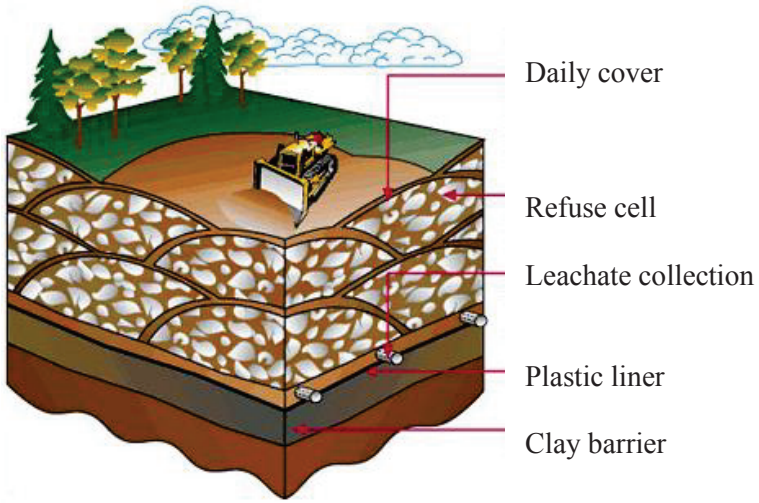
Health Hazards of Landscape Pollution

a. Improper handling of solid wastes. During handling and transfer of biological wastes (from hospitals and clinics) disease transmission may take place by infection through open sores or vectors like rats and insects which invade refuse dumps for food
b. Rats spread many diseases like plague, salmonellosis, endemic typhus, trichinosis, etc. through direct bite. They quickly proliferate and spread to neighboring areas destroying property and spreading diseases. Flies breed on feces etc. from where they migrate to food and water and result in transmission of many diseases like bacillary dysentery, diarrhea and amoebic dysentery in humans.
c. The improper disposal of hazardous wastes results in contamination of crops or water supplies and thus pose a serious health hazard for humans and animals resulting in acute effects like death. Large scale epidemic of cholera, gastro-intestinal diseases, jaundice, and hepatitis etc. result from contamination of soil and water bodies by the leachate from decomposed and purified garbage dumps.
d. Chocking of drains and gully pits by the solid wastes result in water logging especially during the rainy season. This water-logging results in the breeding of mosquitoes in the stagnant water.

Methods of Waste Disposal

Waste should be disposed of properly. Depending on the location, there are generally five techniques used in waste disposal. (Olivar, 2016)

1. *Landfills* - Landfill method is the cheapest and the most convenient waste disposal. A typical solid waste landfill is constructed above an impermeable clay layer that is lined with an impermeable membrane. The everyday deposit of fresh garbage is covered with a layer of oil to prevent it from blowing around. The threat of groundwater contamination reduces the use of this practice.



Reference: http://www.bra.org/landfills.html

The figure shows the part of a compost pit which are as follows:

- *Daily cover* - no landfill refuse is left exposed overnight- at the end of each day, all refuse is covered with at least six (6) inches of compacted soil.
- *Refuse cell* - compacted garbage surrounded by soil from daily cover
- *Leachate collection* - perforated pipes in a layer of sand collect rainwater that has filtered through the landfill.
- *Plastic liner* - prevents soil and water contamination
- *Clay barrier*- prevents soil and water contamination

2. *Incineration* - This is burning in a controlled manner using an incinerator. Incineration coupled with high temperature waste treatments are recognized as thermal treatments. During the process of incineration, the waste material that is treated is converted into gases, particles, and heat. These products are later used for generation of electricity. The gases are first treated for eradication of pollutants before going in to atmosphere.

3. *Mulch and Compost* - The organic material that is used to cover the soil is called mulch. Composting is the method of allowing the natural process of decomposition to transform organic materials into humus-like material called compost.



Reference: http://www.treesthatpleasenurseryblog.com/2012/01/got-mulch.html

4. *Source reduction* - This refers to the method of designing, manufacturing, purchasing, suing, and reusing materials so that the amount of waste or its toxicity is reduced.



Remember: Reducing the amount of throwaway food also reduces waste

Reference: http://www.ecolivingadvice.com/7-practical-steps-to-reduce-the-food-you-throw-away/

5. *Recycling* - This is the method of collecting and processing materials that would otherwise be thrown away as trash and turning them into new products.

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